

QUEST DIAGNOSTICS LABORATORY

NORTHSIDE CLINICAL PARTNER | 100 MEDICAL PLAZA, SEATTLE, WA

PATIENT DEMOGRAPHICS	ORDER SUMMARY
Name: Alex Rivera	Lab Order ID: ORD-992182
Age / Gender: 34 / Non-binary	Date Collected: June 18, 2026 08:30 AM
Known Allergies: Penicillin, Strawberries	Date Reported: June 18, 2026 02:45 PM
Primary Physician: Dr. Claire Finch, MD	Clinic Account: Northside Family Care

- CRITICAL IMMEDIATE ATTENTIONS:
- Marked Leukocytosis (Elevated WBC):** Current white blood cell value is **14.2 x10³/uL**, reflecting active inflammatory response.
 - Neutrophilia (82%):** High relative count indicates acute bacterial defense signaling.

1. Complete Blood Count (CBC) with Differential

Biomarker Parameter	Patient Value	Flag Status	Reference Range	Clinical Metric
White Blood Cells (WBC)	14.2	HIGH [A]	4.5 – 11.0	x10 ³ /uL
Red Blood Cells (RBC)	4.8	Normal	4.3 – 5.9	x10 ⁶ /uL
Hemoglobin (Hgb)	15.1	Normal	13.5 – 17.5	g/dL
Hematocrit (Hct)	44.5	Normal	41.0 – 50.0	%
Platelets	350	Normal	150 – 450	x10 ³ /uL
Neutrophils	82	HIGH [A]	40 – 70	%
Lymphocytes	12	LOW [A]	20 – 40	%
Monocytes	5.2	Normal	2.0 – 10.0	%

Note Flag Key: [A]= Critical Out of Bounds. Normal ranges derived from default adult regional benchmarks.

CLINICAL METRICS & COPAY AUDIT

2. Baseline Metabolic Panel Summary

Metabolic Indicator	Patient Value	Flag Status	Reference Range	Clinical Metric
Serum Glucose (Fasting)	94	Normal	70 – 99	mg/dL
Blood Urea Nitrogen (BUN)	16	Normal	7 – 20	mg/dL
Serum Creatinine	0.88	Normal	0.60 – 1.30	mg/dL
Estimated GFR (eGFR)	>90	Normal	>60	mL/min/1.73m ²

Autonomous Clinical Guidance Pathway

The dynamic routing algorithm has verified the biochemical results with the patient’s record:

1. **Suspected Indication:** Significant elevation of total leukocytes ($14.2 \times 10^3/\mu\text{L}$) combined with acute neutrophilia (82%) strongly supports the clinical presentation of an acute bacterial respiratory tract pathway.
2. **Critical Allergy Warning:** Patient memory contains a **high-alert allergy warning to Penicillin**. Traditional treatment profiles (such as Amoxicillin or Augmentin) must be avoided to prevent immediate systemic hypersensitivity. Safe secondary alternatives (e.g., Macrolides like Azithromycin) are suggested for consideration.

RECOMMENDATION QUESTIONS FOR PHYSICIAN DISCUSSION:

1. Given my WBC count of 14.2 and high neutrophils, does this point clearly to an active bacterial infection instead of viral flu?
2. I see Amoxicillin was under discussion. Because of my noted Penicillin allergy, what non-beta-lactam alternative therapies are appropriate for me?
3. Do we need to schedule a follow-up Complete Blood Count (CBC) panel next week to monitor the leukocyte resolution?

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